

Program 10: Demonstrate kubernetes Dashboard and CLI for cluster monitoring

Project Structure:

Pgm10

->deployment.yaml

->service.yaml

STEP 1 - Start Minikube

minikube start

```
lrv24mc058_kumaraswamys@jarvis:~$ minikube start
🐹 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" .../
```

STEP 2 - Check Minikube Status

minikube status

You should see:

host: Running

kubelet: Running

apiserver: Running

kubeconfig: Configured

```
lrv24mc058_kumaraswamys@jarvis:~$ minikube status
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Create a node js app

Nano server.js

```
const express = require('express')
```

```
const app = express()
```

```
const port = 3000
```

```
app.get('/', (req, res) => {
  res.send('Hello World!')
})
```

```
app.listen(port, () => {  
  console.log(`Example app listening on port ${port}`)  
})
```

STEP 3 - If Minikube is not running, restart it

```
minikube delete  
minikube start --driver=docker
```

STEP 4 - Create the Deployment file (nodejs-deployment.yaml)

Create a new file:

```
nginx-deployment.yaml  
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: nodejs-deployment  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: nginxtemplate:  
  metadata:  
    labels:  
      app: nginxspec:  
  containers:  
    - name: node  
      image: node:latest  
  ports:  
    - containerPort: 3000
```

STEP 5 - Apply the deployment

```
kubectl apply -f node-deployment.yaml
```

STEP 6 - Check Deployment & Pods

```
kubectl get deployments  
kubectl get pods  
You should see 2 pods running.
```

STEP 7 -Expose the Deployment as a Service

```
kubectl expose deployment node-deployment \  
--type=NodePort \  
--port=80  
This creates a service that exposes node to your system.
```

STEP 8 - Check Services

```
kubectl get services
```

```

lrvt24mc058_kumaraswamys@jarvis:~$ kubectl get services
NAME                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes           ClusterIP     10.96.0.1     443/TCP        8m12s
node-deployment      NodePort      10.101.24.25  80:30269/TCP   11s

```

A service named node-deployment should appear .

STEP 9 - Access the Webpage

Minikube provides a shortcut to open the app:minikube service nginx-deployment

This will open the browser showing the Nginx Welcome Page.

STEP 10 — Clean Up

Delete all pods

`kubectl delete pods --all`

Delete all services

`kubectl delete svc --all`

Delete all deployments

`kubectl delete deployments --all`



Open Kubernetes Dashboard

```

lrvt24mc058_kumaraswamys@jarvis:~$ minikube dashboard
🚀 Verifying dashboard health ...
🔧 Launching proxy ...
😊 Verifying proxy health ...
🌐 Opening http://127.0.0.1:43167/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-
dashboard:/proxy/ in your default browser...
Opening in existing browser session.

```

kubernetes

default

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Running 1

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Deployments

Name	Images	Labels	Pods	Created
nginx-deployment	nginx:latest	app: nginx	2 / 2	an hour ago

Pods

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
program8	node:18-alpine	app: nodejs	minikube	Running	0	-	-	36 minutes ago
nginx-deployment-6f966444b-8pvr	nginx:latest	app: nginx pod-template-hash: 6f966444b	minikube	Running	0	-	-	an hour ago
nginx-deployment-6f966444b-z9js	nginx:latest	app: nginx pod-template-hash: 6f966444b	minikube	Running	0	-	-	an hour ago